Sports Science Research Update

January 2023

A delayed evening meal enhances sleep quality in young rugby players. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-8, 2022.

Adductor strengthening program is successfully adopted but frequently modified in Norwegian male professional soccer teams: A cross-sectional study. BMJ Open, 12(e060611): 1-8, 2022.

Are EPB41 and alpha-synuclein diagnostic biomarkers of sport-related concussion? Findings from the NCAA and Department of Defense CARE Consortium. Journal of Sport and Health Science, Ahead of Print: 1-22, 2022.

Comparison of eccentric hamstring strength and asymmetry at return-to-sport after hamstring strain injury among those who did and did not re-injure. Physical Therapy in Sport, 59: 25-29, 2023.

Effect of mindfulness-based programs on elite athlete mental health: A systematic review and meta-analysis. British Journal of Sports Medicine, Ahead of Print: 1-12, 2022.

Effects of an external pneumatic compression device vs static compression garment on peripheral circulation and markers of sports performance and recovery. European Journal of Applied Physiology, 122: 1709-1722, 2022.

Effects of fatigue induced by repeated sprints on sprint biomechanics in football players: Should we look at the group or the individual? International Journal of Environmental Research and Public Health, 19(14643): 1-12, 2022.

Evaluating the relationship between neurological function, neuromuscular fatigue, and subjective performance measures in professional rugby union players. Frontiers in Sports and Active Living, Nov. 11: 1-8, 2022.

Genetic profile in genes associated with muscle injuries and injury etiology in professional soccer players. Frontiers in Genetics, Nov. 16: 1-15, 2022.

Sports Science Research Update

Incidence and prevalence of hamstring injuries in field-based team sports: A systematic review and meta-analysis of 5952 injuries from over 7 million exposure hours. British Journal of Sport Medicine, Ahead of Print: 1-9, 2022.

Investigating the sleep habits in individual and team-sport athletes using the Athlete Sleep Behavior Questionnaire and the Pittsburgh Sleep Quality Index. Sleep Science, 15(1): 112-117, 2022.

Killing two birds with one stone: Evaluate sprinting horizontal force production capacities to improve performance and injury risk management. Blog: British Journal of Sports Medicine, Nov. 11: 1-6, 2022.

Low prevalence of cardiac abnormalities in competitive athletes at return to play after COVID-19. Journal of Science and Medicine in Sport, Ahead of Print: 1-6, 2022.

Neurocognitive outcomes of older National Football League retirees. Brain Injury, Ahead of Print: 1-9, 2022.

Placebo response and media attention in randomized clinical trials assessing cannabis-based therapies for pain: A systematic review and meta-analysis. JAMA Network Open, 11(e2243848): 1-12, 2022.

Sleep architecture in response to a late evening competition in team-sport athletes. International Journal of Sports Physiology and Performance, 17: 569-575, 2022.

Synthetic peptides in doping control: A powerful tool for an analytical challenge. ACS Omega, 7: 38193-38206, 2022.

Vibration Therapy – A clinical commentary. International Journal of Sports Physical Therapy, 17(6): 984-987, 2022.

Sports Science Research Update

A delayed evening meal enhances sleep quality in young rugby players. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-8, 2022.

Abstract - The aim of this study was to examine the effect of delayed evening mealtime on sleep quality in young athletes. Twelve rugby players (age 15.8 ± 0.7 years) participated in a crossover within-participant design. Adolescents spent five consecutive days in each of two conditions, separated by a 2-week washout period: routine dinner (3.5 hr before bedtime) and late dinner (LD, 1.5 hr before bedtime). Other mealtimes as well as bedtime and wake-up time were usual and remained the same in both conditions. Their schedules, dietary intakes, and physical activity were controlled and kept constant throughout the study. Sleep was assessed using polysomnography on the first and the last nights in the individual rooms of the boarding school. An increase in total sleep time by 24 min ($p = .001$, $d = 1.24$) and sleep efficiency by 4.8% was obtained during LD ($p = .001$, $d = 1.24$). Improvement in sleep efficiency was mainly due to a lower wake after sleep onset (−25 min, $p = .014$, $d = -3.20$), a decrease of microarousals (−25%, $p = .049$, $d = -0.64$), and awakenings ≥90 s (−30%, $p < .01$, $d = -0.97$) in LD compared to routine dinner. There were no significant differences in sleep architecture except for a shorter slow-wave sleep (N3) latency (−6.9 min, $p = .03$, $d = -0.778$) obtained during LD. In this study, evening dinner 1.5 hr before bedtime leads to better quality and less fragmented sleep compared to evening dinner 3.5 hr before bedtime in young athletes.

Sports Science Research Update

Adductor strengthening program is successfully adopted but frequently modified in Norwegian male professional soccer teams: A cross-sectional study. BMJ Open, 12(e060611): 1-8, 2022.

Objectives: Groin injuries represent a substantial problem in male football, with the Adductor Strengthening Program (ASP) being the only exercise program demonstrated to significantly reduce the risk of groin problems. We aimed first, to use the Reach Adoption Effectiveness Implementation Maintenance (RE-AIM) framework to investigate attitudes, beliefs and behavior to the ASP among primary delivery agents of injury prevention exercises in Norwegian male professional football teams. Second, we aimed to identify a real-world application of the ASP protocol used in a professional team setting.

Design: A descriptive cross-sectional survey, using a questionnaire designed to cover all five dimensions of the RE-AIM framework.

Setting: The top two divisions of Norwegian male professional football.

Participants: 32 primary injury prevention delivery agents.

Primary and secondary outcome measures: Primarily, the proportion of respondents being aware of the ASP and its effect; having adopted it; having implemented it as intended; and considering maintaining using it. Secondary, the most often used ASP modifications.

Results: Twenty-nine (91%) participants responded. All (100%) respondents were aware of the ASP and its injury preventive effect. The two most stated reasons for using the ASP were its injury preventive effect and that it does not require equipment. The ASP was adopted by all (100%) delivery agents, but only 10% used it in accordance with the original protocol. The main modifications were that the players in 72% of the teams were instructed to perform a non-progressive number of repetitions during pre-season, and in 86% of the teams instructed to perform more sets, but fewer repetitions per set, during in season. In total, 97% of the delivery agents planned to continue using the ASP.

Conclusion: The delivery agents have positive attitudes and beliefs to the ASP, but they frequently modify it. We identified and reported a real-world application of the ASP protocol.

Strengths and limitations of the Study

⇒ The questionnaire was pilot tested by delivery agents with relevant experience.

⇒ Thorough data collection process leading to a high response rate.

⇒ The internal validity of the questionnaire was not systematically explored.

⇒ Some of the questionnaire's questions are prone to recall bias as the survey was conducted towards the end of the competitive season.

Sports Science Research Update

Are EPB41 and alpha-synuclein diagnostic biomarkers of sport-related concussion? Findings from the NCAA and Department of Defense CARE Consortium. Journal of Sport and Health Science, Ahead of Print: 1-22, 2022.

Background: Current protein biomarkers are only moderately predictive at identifying individuals with mild traumatic brain injury or concussion. Therefore, more accurate diagnostic markers are needed for sport-related concussion (SRC).

Methods: This was a multicenter, prospective, case-control study of athletes who provided blood samples and were diagnosed with a concussion or were a matched non-concussed control within the NCAA–DoD Concussion Assessment, Research, and Education (CARE) Consortium conducted between 2015 and 2019. The blood was collected within 48 h of injury to identify protein abnormalities at the acute and subacute timepoints. Athletes with concussion were divided into 6 h post-injury (0–6 h post-injury) and after 6 h post-injury (7–48 hours post-injury) groups. We applied a highly multiplexed proteomic technique that used a DNA aptamers assay to target 1305 proteins in plasma samples from athletes with and without SRC.

Results: A total of 140 athletes with concussion (79.3% male; aged 18.71 ± 1.10 years, mean ± SD) and 21 non-concussed athletes (76.2% male; 19.14 ± 1.10 years) were included in this study. We identified 338 plasma proteins that significantly differed in abundance (319 upregulated and 19 downregulated) in concussed athletes compared to non-concussed athletes. The top 20 most 4 differentially abundant proteins discriminated concussed athletes from non-concussed athletes with an area under the curve (AUC) of 0.954 (95% confidence interval: 0.922–0.986). Specifically, after 6 h of injury, the individual AUC of plasma erythrocyte membrane protein band 4.1 (EPB41) and alpha-synuclein (SNCA) were 0.956 and 0.875, respectively. The combination of EPB41 and SNCA provided the best AUC (1.000), which suggests this combination of candidate plasma biomarkers is best for diagnosing concussion in athletes after 6 h of injury.

Conclusion: Our data suggest that proteomic profiling may provide novel diagnostic protein markers and that a combination of EPB41 and SNCA is the most predictive biomarker of concussion after 6 h of injury.

Highlights

- Feasibility of multiplexed blood-based protein levels in sport-related concussion.
- Identify novel blood-based biomarkers for concussed athletes.
- Dysregulation of blood-based proteins provides a better understanding of human response to injury.
- EPB41 and alpha synuclein distinguish concussed athletes from non-concussed athletes.

Sports Science Research Update

Comparison of eccentric hamstring strength and asymmetry at return-to-sport after hamstring strain injury among those who did and did not re-injure. Physical Therapy in Sport, 59: 25-29, 2023.

Objectives: The primary aim of this study was to describe eccentric hamstring strength magnitude and asymmetry at the time of return-to-sport (RTS) after an index hamstring strain injury (HSI) and determine if there were differences in strength asymmetry at RTS between those who did and did not go on to re-injure within 1-month and within 3-months of RTS.

Design: Cross-sectional study. Setting: Laboratory-based.

Participants: Sixty National Collegiate Athletic Association Division I athletes with index HSI.

Main Outcome Measures: Maximum hamstring eccentric strength for each limb, total maximum strength summed across limbs, and between-limb asymmetry at the time of RTS following the index HSI, assessed using the NordBord Hamstring Testing System.

Results: Of the 60 index HSIs, 8 (13%), and 11 (18%) re-injuries occurred within 1 and 3-months of RTS, respectively. There were no differences between those who did and did not re-injure in maximum eccentric force of either limb (p-values ≥ 0.52), total force from both limbs (p-values ≥ 0.47), and between limb force asymmetry (p-values ≥ 0.91), regardless if re-injury occurred within 1 or 3-months after RTS.

Conclusions: Eccentric hamstring strength and asymmetry measured at the time of RTS did not differ between those who did and did not re-injure within 3-months of RTS.

Sports Science Research Update

Effect of mindfulness-based programs on elite athlete mental health: A systematic review and meta-analysis. British Journal of Sports Medicine, Ahead of Print: 1-12, 2022.

Objective: To determine the effectiveness of mindfulness-based programs (MBPs) on the mental health of elite athletes.

Design: Systematic review and meta-analysis.

Data sources: **Eight online databases (Embase, PsycINFO, SPORTDiscus, MEDLINE, Scopus, Cochrane** CENTRAL, ProQuest Dissertations & Theses and Google Scholar), plus forward and backward searching from included studies and previous systematic reviews.

Eligibility criteria for selecting studies: Studies were included if they were randomized controlled trials (RCTs) that compared an MBP against a control, in current or former elite athletes.

Results: Of 2386 articles identified, 12 RCTs were included in this systematic review and meta-analysis, comprising a total of 614 elite athletes (314 MBPs and 300 controls). Overall, MBPs improved mental health, with large significant pooled effect sizes for reducing symptoms of anxiety (hedges g=−0.87, number of studies (n)=6, p=0.017, I 2 =90) and stress (g=−0.91, n=5, p=0.012, I 2 =74) and increasing psychological well-being (g=0.96, n=5, p=0.039., I 2 =89). Overall, the risk of bias and certainty of evidence was moderate, and all findings were subject to high estimated levels of heterogeneity.

Conclusion: MBPs improved several mental health outcomes. Given the moderate degree of evidence, high-quality, adequately powered trials are required in the future. These studies should emphasize intervention fidelity, teacher competence and scalability within elite sport.

Key messages

What is already known?

⇒ Mindfulness-based programs (MBPs) are comparable to other interventions to treat mental health symptoms and disorders in the general population.

⇒ MBPs are associated with improvements in athletic performance.

⇒ Athlete engagement with psychological interventions to improve mental health might be greater with interventions that also support performance.

What are the new findings?

Sports Science Research Update

⇒ This is the first systematic review and meta-analysis of MBPs that focused on mental health outcomes exclusively in elite athletes, evaluating evidence from randomized controlled trials only.

⇒ MBPs are an acceptable form of intervention in elite sport settings and show promise in improving several mental health outcomes including stress, anxiety and psychological well-being.

⇒ Adapted MBPs that teach mindfulness in a sport-specific manner may be more effective for treating mental health symptoms in elite athletes.

⇒ MBPs may facilitate performance alongside mental health.

How might this study impact clinical practice in the future?

⇒ MBPs should be adapted to the sporting and social context.

⇒ Core elements of a high-fidelity MBP should be retained, particularly when targeting specific mental health symptoms or disorders.

⇒ Framing the MBP as performance support may decrease the stigma associated with help seeking for mental health symptoms and increase engagement with the program.

⇒ MBPs may reduce symptoms of anxiety and competition anxiety, highlighting the value of MBPs to support performance alongside mental health.

Sports Science Research Update

Effects of an external pneumatic compression device vs static compression garment on peripheral circulation and markers of sports performance and recovery. European Journal of Applied Physiology, 122: 1709-1722, 2022.

Purpose: To identify the effects of a single 30 min partial lower leg external pneumatic compression (EPC) treatment compared to a static compression (SC) garment or a no treatment control (CTL) on markers of recovery and performance following a muscle damaging protocol.

Methods: Thirty healthy, active males (23±3 years; 180.2±9.0 cm; 81.6±11.3 kg) performed 100 drop jumps from a 0.6 m box followed by a randomized, single 30 min treatment of either a partial lower leg EPC device worn below the knee and above the ankle (110 mmHg), SC garment (20–30 mmHg) covering the foot and calf just below the knee, or no treatment CTL, and then returned 24 and 48 h later. Participants were assessed for measures of muscle soreness, fatigue, hemodynamics, blood lactate, muscle thickness, circumferences, and performance assessments.

Results: The drop jump protocol significantly increased muscle soreness ($p<0.001$), fatigue ($p<0.001$), blood flow ($p<0.001$), hemoglobin ($p<0.001$), and muscle oxygen saturation (SMO2; $p<0.001$). Countermovement jump and squat jump testing completed after treatment with either EPC, SC, or CTL revealed no differences for jump height between any condition. However, EPC treatment maintained consistent braking force and propulsive power measures across all timepoints for countermovement jump testing. EPC and SC treatment also led to better maintenance of squat jump performance for average relative propulsive force and power variables at 24 and 48 h compared to CTL.

Conclusions: A single 30 min partial leg EPC treatment may lead to more consistent jump performance following a damaging bout of exercise.

Sports Science Research Update

Effects of fatigue induced by repeated sprints on sprint biomechanics in football players: Should we look at the group or the individual? International Journal of Environmental Research and Public Health, 19(14643): 1-12, 2022.

Abstract - The aim of this study was to analyze the influence of fatigue on sprint biomechanics. Fifty-one football players performed twelve maximal 30 m sprints with 20 s recovery between each sprint. Sprint kinetics were computed from running speed data and a high-frequency camera (240 Hz) was used to study kinematic data. A cluster analysis (K-mean clustering) was conducted to classify individual kinematic adaptations. A large decrease in maximal power output and less efficiency in horizontally orienting the ground reaction force were observed in fatigued participants. In addition, individual changes in kinematic components were observed, and, according to the cluster analysis, five clusters were identified. Changes in trunk, knee, and hip angles led to an overall theoretical increase in hamstring strain for some players (Cluster 5, 20/51) but to an overall decrease for some others (Cluster 1, 11/51). This study showed that the repeated sprint ability (RSA) protocol had an impact on both kinetics and kinematics. Moreover, fatigue affected the kinematics in a different way for each player, and these individual changes were associated with either higher or lower hamstring length and thus strain.

Sports Science Research Update

Evaluating the relationship between neurological function, neuromuscular fatigue, and subjective performance measures in professional rugby union players. Frontiers in Sports and Active Living, Nov. 11: 1-8, 2022.

Abstract - Managing the health and wellbeing of full-time professional athletes is a multifaceted task. In elite high-performance environments, medical staff and strength training coaches attempt to identify improved methods to monitor player health. Monitoring player health could indicate potential injury risk and assist in adjustments to training and workload management. Measuring fatigue is a notable component of monitoring player readiness before and after training sessions, and after competitive fixtures. In the present study, a novel method of gathering non-invasive player data was investigated by utilizing the Omegawave (OW) to monitor direct current (DC) potential brainwave activity. This method allowed for non-invasive data gathering to assess recovery, player readiness and indicators of workload that may affect optimal performance. DC potential is based on recording low electrical frequencies (>0.5 Hz) that is derived from (1) Stabilization point of DC potential (mV), (2) Stabilization time (1.0–7.0) and (3) Curve shape (1.0–7.0). These measures evaluate the athlete's internal stress, readiness to perform, and neurological function through DC potential brain wave activity and heart rate variability (HRV) assessments. The primary aim of this case series was to compare the efficacy of objective DC potential brainwave activity measurements (neurological function) with neuromuscular fatigue data using reactive strength index modified (RSImod) and profile of mood states (POMS) questionnaires to assess a player's state of readiness to train. The participants in this study were eleven male senior professional rugby union players with a mean stature (±SD) of 185.2 ± 8.6 cm, mass of 101.1 ± 12.9 kg, and age of 27.1 ± 2.1 years. All players were tested 3 days per week over a 6-week mid-season period. Results from this case study suggest that DC potentials could be used as an objective measure to indicate player readiness and managing individual player workload. The final analyses identified a weak negative correlation ($r = -0.17$) between the RSImod data and the DC potential data was observed. DC potential brainwave activity data could be used in conjunction with subjective measures such as POMS, RSImod and reported injury status to adjust player daily activity.

Sports Science Research Update

Genetic profile in genes associated with muscle injuries and injury etiology in professional soccer players. Frontiers in Genetics, Nov. 16: 1-15, 2022.

Abstract - Many causes define injuries in professional soccer players. In recent years, the study of genetics in association with injuries has been of great interest. The purpose of this study was to examine the relationship between muscle injury-related genes, injury risk and injury etiology in professional soccer players. In a cross-sectional cohort study, one hundred and twenty-two male professional football players were recruited. AMPD1 (rs17602729), ACE (rs4646994), ACTN3 (rs1815739), CKM (rs8111989) and MLCK (rs2849757 and rs2700352) polymorphisms were genotyped by using Single Nucleotide Primer Extension (SNPE). The combined influence of the six polymorphisms studied was calculated using a total genotype score (TGS). A genotype score (GS) of 2 was assigned to the "protective" genotype for injuries, a GS of 1 was assigned to the heterozygous genotype while a GS of 0 was assigned to the "worst" genotype. Injury characteristics and etiology during the 2021/2022 season were classified following a Consensus Statement for injuries recording. The distribution of allelic frequencies in the AMPD1 and MLCK c.37885C>A polymorphisms were different between non-injured and injured soccer players ($p < 0.001$ and $p = 0.003$, respectively). The mean total genotype score (TGS) in non-injured soccer players (57.18 ± 14.43 arbitrary units [a.u.]) was different from that of injured soccer players (51.71 ± 12.82 a.u., $p = 0.034$). There was a TGS cut-off point (45.83 a.u.) to discriminate noninjured from injured soccer players. Players with a TGS beyond this cut-off had an odds ratio of 1.91 (95%CI: 1.14–2.91; $p = 0.022$) to suffer an injury when compared with players with lower TGS. In conclusion, TGS analysis in muscle injury-related genes presented a relationship with professional soccer players at increased risk of injury. Future studies will help to develop this TGS as a potential tool to predict injury risk and perform prevention methodology in this cohort of football players.

Sports Science Research Update

Incidence and prevalence of hamstring injuries in field-based team sports: A systematic review and meta-analysis of 5952 injuries from over 7 million exposure hours. British Journal of Sport Medicine, Ahead of Print: 1-9, 2022.

Objective: This study aimed to systematically review and meta-analyze the incidence and prevalence of hamstring injuries in field-based team sports. A secondary aim was to determine the impact of other potential effect moderators (match vs training; sport; playing surface; cohort age, mass and stature; and year when data was collected) on the incidence of hamstring injury in field-based team sports.

Design: Systematic review and meta-analysis.

Data sources: CINAHL, Cochrane Library, MEDLINE Complete (EBSCO), Embase, Web of Science and SPORTDiscus databases were searched from database inception to 5 August 2020.

Eligibility criteria: Prospective cohort studies that assessed the incidence of hamstring injuries in field-based team sports.

Method: Following database search, article retrieval and title and abstract screening, articles were assessed for eligibility against predefined criteria then assessed for methodological quality using the Critical Appraisal Tool for prevalence studies. Meta-analysis was used to pool data across studies, with meta-regression used where possible.

Results: Sixty-three articles were included in the meta-analysis, encompassing 5952 injuries and 7 262 168 hours of exposure across six field-based team sports (soccer, rugby union, field hockey, Gaelic football, hurling and Australian football). Hamstring injury incidence was 0.81 per 1000 hours, representing 10% of all injuries. Prevalence for a 9-month period was 13%, increasing 1.13-fold for every additional month of observation ($p=0.004$). Hamstring injury incidence increased 6.4% for every 1 year of increased average cohort age, was 9.4-fold higher in match compared with training scenarios ($p=0.003$) and was 1.5-fold higher on grass compared with artificial turf surfaces ($p<0.001$). Hamstring injury incidence was not significantly moderated by average cohort mass ($p=0.542$) or stature ($p=0.593$), was not significantly different between sports ($p=0.150$) and has not significantly changed over the last 30 years ($p=0.269$).

Sports Science Research Update

Conclusion: Hamstring injury represents 10% of all injuries in field-based team sports, with 13% of the athletes experiencing a hamstring injury over a 9-month period most commonly during matches. More work is needed to reduce the incidence of hamstring injury in field-based team sports.

What are the findings?

⇒ The incidence of hamstring injury in field-based team sports is 0.81 per 1000 exposure hours, while the prevalence is 13% for a 9-month period, increasing 1.13-fold for every subsequent month of observation.

⇒ Hamstring injury incidence has not changed over the last 30 years.

⇒ The incidence of hamstring injuries is moderated by non-modifiable and modifiable factors, with matches, increasing athlete age and grass surfaces (compared with artificial turf) associated with a greater incidence of hamstring injury.

How might it impact on clinical practice in the future?

⇒ Clinicians can expect a higher incidence of hamstring injuries in older athletes compared to younger athletes.

⇒ Clinicians can expect a higher incidence of hamstring injuries during matches compared to training.

⇒ Clinicians should consider the potential for higher incidence of hamstring injuries on grass surfaces compared to artificial turf during soccer.

Sports Science Research Update

Investigating the sleep habits in individual and team-sport athletes using the Athlete Sleep Behavior Questionnaire and the Pittsburgh Sleep Quality Index. Sleep Science, 15(1): 112-117, 2022.

Objectives: There is limited large-scale data on maladaptive sleep practices in elite adult athletes and their influence on sleep characteristics. This study aimed to identify differences in sleep behaviors between individual and team-sport athletes using two sleep questionnaires.

Material and Methods: 407 (237 male, 170 female) elite adult athletes across sixteen sports (9 individual-sports, 7 team-sports) completed the Athlete Sleep Behavior Questionnaire (ASBQ) and the Pittsburgh Sleep Quality Index (PSQI).

Results: Individual-sport athletes reported greater total sleep time and higher sleep efficiency than team-sport athletes ($p<0.05$, $d=0.28$-0.29). There were no differences between global scores for the PSQI, however, there was a difference between global scores for the ASBQ as well as multiple individual items from both questionnaires ($p<0.05$), indicating poorer behaviors in team-sport athletes.

Discussion: Team-sport athletes displayed more maladaptive pre-sleep behaviors and poorer sleep characteristics than individual-sport athletes.

Sports Science Research Update

Killing two birds with one stone: Evaluate sprinting horizontal force production capacities to improve performance and injury risk management. Blog: British Journal of Sports Medicine, Nov. 11: 1-6, 2022.

Overview: Sprinting is important for athletes, coaches, and health professionals for both performance and injury risk management [1]. Sprinting can be defined as running with the intention of maximal acceleration and/or velocity. The performance of sprinting acceleration is associated with the ability to produce and apply high levels of force in the horizontal direction [2]. Such information is of great importance when we want to evaluate, monitor and/or train for sprinting. Evaluating and monitoring sprinting horizontal force production capacities is now used in field settings for sprinting performance enhancement, [3,4] as well as in more analytical evaluations such as isolated single joint strength assessments. This ability to produce horizontal force during sprinting is mainly associated with the lower limb posterior chain muscles actions [5,6]. Thus, we can consider that sprinting horizontal force production capacities evaluation allows an overall evaluation of the lower limb function during sprinting action. More specifically, this evaluation targets the overall posterior chain muscles function during sprinting action. Recent studies also supported the interest of evaluating sprinting horizontal force production for hamstring injury management [7–11]. This blog article will explain how to assess sprinting horizontal force production capacities and why they can be of interest for both injury risk management and to performance improvement.

Sports Science Research Update

Low prevalence of cardiac abnormalities in competitive athletes at return to play after COVID-19. Journal of Science and Medicine in Sport, Ahead of Print: 1-6, 2022.

Objective: To evaluate the prevalence of cardiac involvement after COVID-19 in competitive athletes at return-to-play (RTP) evaluation, following the recommended Italian protocol including cardiopulmonary exercise test (CPET) and 24-Hour Holter monitoring.

Design and methods: This is a single center observational, cross-sectional study. Since October 2020, all competitive athletes (age ≥ 14 years) evaluated in our Institute after COVID-19, prior RTP were enrolled. The protocol dictated by the Italian governing bodies included: 12-lead ECG, blood test, CPET, 24-h ECG monitoring, spirometry. Cardiovascular Magnetic Resonance (CMR) was performed based on clinical indication.

Results: 219 consecutive athletes were examined (59% male), age 23 years (IQR 19–27), 21% asymptomatic, 77% mildly symptomatic, 2% with previous pneumonia. The evaluation was performed after a median of 10 (6–17) days from negative SARS-CoV-2 swab. All athletes showed a good exercise capacity at CPET without cardiovascular and respiratory limitations. Uncommon premature ventricular contractions (PVCs) were found in 9.5% (n = 21) at CPET/Holter ECG monitoring. Two athletes (0.9%) were diagnosed with acute myocarditis (by CMR) and another one with new pericardial effusion. All the three athletes were temporally restricted from sport participation.

Conclusions: Myocarditis in competitive athletes screened after COVID-19 resolution was detected in a low minority of the cases (0.9%). However, a non-negligible prevalence of uncommon PVCs (9%) was observed, either at CPET and/or Holter ECG monitoring, including all athletes with COVID-19 related cardiovascular abnormalities.

Sports Science Research Update

Neurocognitive outcomes of older National Football League retirees. Brain Injury, Ahead of Print: 1-9, 2022.

Objective: Determine if head-injury exposure relates to later-in-life cognitive decline in older National Football League (NFL) retirees.

Method: NFL retirees (aged 50+) with or without cognitive impairment underwent baseline (n = 53) and follow-up (n = 29; 13–59 months later) neuropsychological evaluations. Cognitively normal (CN) retirees (n = 26) were age- and education-matched to healthy controls (n = 26). Cognitively impaired (CI) retirees with mild cognitive impairment or dementia (n = 27) were matched to a clinical sample (CS) by age, sex, education, and diagnosis (n = 83). ANOVAs compared neuropsychological composites at baseline and over time between retirees and their matched groups. Regression models evaluated whether concussions, concussions with loss of consciousness (LOC), or games played predicted neuropsychological functioning.

Results: At baseline, CN retirees had slightly worse memory than controls ($M_{CN\ retirees} = 50.69$, $SE_{CN\ retirees} = 1.320$; $M_{Healthy}$ controls = 57.08, $SE_{Healthy}$ controls = 1.345; $p = 0.005$). No other group differences were observed, and head-injury exposure did not predict neurocognitive performance at baseline or over time.

Conclusions: Head-injury exposure was not associated with later-in-life cognition, regardless of cognitive diagnosis. Some retirees may exhibit lower memory scores compared to age-matched peers, though this is of unclear clinical significance.

Sports Science Research Update

Placebo response and media attention in randomized clinical trials assessing cannabis-based therapies for pain: A systematic review and meta-analysis. JAMA Network Open, 11(e2243848): 1-12, 2022.

Importance: Persistent pain is a common and disabling health problem that is often difficult to treat. There is an increasing interest in medicinal cannabis for treatment of persistent pain; however, the limited superiority of cannabinoids over placebo in clinical trials suggests that positive expectations may contribute to the improvements.

Objective: To evaluate the size of placebo responses in randomized clinical trials in which cannabinoids were compared with placebo in the treatment of pain and to correlate these responses to objective estimates of media attention.

Data Sources: A systematic literature search was conducted within the MEDLINE and Embase databases. Studies published until September 2021 were considered.

Study Selection: Cannabinoid studies with a double-blind, placebo-controlled design with participants 18 years or older with clinical pain of any duration were included. Studies were excluded if they treated individuals with HIV/AIDS or severe skin disorders.

Data Extraction and Synthesis: The study followed the Preferred Reporting Items for Systematic Review and Meta-analyses reporting guideline. Data were extracted by independent reviewers. Quality assessment was performed using the Risk of Bias 2 tool. Attention and dissemination metrics for each trial were extracted from Altmetric and Crossref. Data were pooled and analyzed using a random-effects statistical model.

Main Outcomes and Measures: Change in pain intensity from before to after treatment, measured as bias-corrected standardized mean difference (Hedges g).

Results: Twenty studies, including 1459 individuals (mean [SD] age, 51 [7] years; age range, 33-62 years; 815 female [56%]), were included. Pain intensity was associated with a significant reduction in response to placebo, with a moderate to large effect size (mean [SE] Hedges g, 0.64 [0.13]; $P < .001$). Trials with low risk of bias had greater placebo responses ($q1 = 5.47$; $I\ 2 = 87.08$; $P = .02$). The amount of media attention and dissemination linked to each trial was proportionally high, with a strong positive bias, but was not associated with the clinical outcomes.

Conclusions and Relevance: Placebo contributes significantly to pain reduction seen in cannabinoid clinical trials. The positive media attention and wide dissemination may uphold high expectations and

shape placebo responses in future trials, which has the potential to affect the outcome of clinical trials, regulatory decisions, clinical practice, and ultimately patient access to cannabinoids for pain relief.

Key Points

Question What is the size of the placebo response in cannabinoid trials for clinical pain, and is the magnitude of placebo response associated with media attention on the trials?

Findings This meta-analysis of 20 studies of 1459 individuals found a significant pain reduction in response to placebo in cannabinoid randomized clinical trials. Media attention was proportionally high, with a strong positive bias, yet not associated with the clinical outcomes.

Meaning These findings suggest that placebo has a significant association with pain reduction as seen in cannabinoid clinical trials, and the positive media attention may shape placebo responses in future trials.

Sports Science Research Update

Sleep architecture in response to a late evening competition in team-sport athletes. International Journal of Sports Physiology and Performance, 17: 569-575, 2022.

Purpose: The aim of this study was to assess the effect of a late evening competition on sleep characteristics and perceived recovery in team-sport players using a validated self-applicable and portable polysomnographic device.

Methods: Sixteen team-sport players (age: 25.4 [1.4] y; body mass index: 23.6 [0.5] kg/m^2) completed the study. Objective and subjective sleep data were collected for 4 consecutive nights: 2 nights before (PRE2 and PRE1) and the 2 nights after an evening match (POST1), performed between 6:00 PM and 9:00 PM. Total quality of recovery values were also collected in the morning.

Results: A significantly delayed bedtime ($P < .0001$; $\eta 2$ p = .68, large) was observed in the first night after the competition (2 h 29 min [1 h 15 min]) compared both to PRE2 (+88 min; $P < .0001$), PRE1 (+98 min; $P < .0001$), and POST1 (+100 min; $P < .0001$), and similar results were observed for wake-up time ($P = .033$; $\eta^2_p = .39$, large): Players woke up significantly later in evening match (9 h 20 min [1 h 55 min]) compared with PRE2 (+85 min; $P = .050$) and POST1 (+85 min; $P = .049$). Conversely, total sleep time; sleep efficiency; sleep onset latency; wake after sleep onset; cortical arousals; N1, N2, N3, and REM (rapid eye movement) percentages; total quality of recovery values; and scores of subjective sleep quality did not vary among the 4 study nights.

Conclusions: Team-sport players had delayed bedtime and wake-up time following an evening competition; however, sleep quality, duration, and subjective scores of recovery were not affected by the evening match. The delayed wake-up time seems to protect athletes' sleep efficiency/duration against the evening-match-induced delayed bedtime.

Sports Science Research Update

Synthetic peptides in doping control: A powerful tool for an analytical challenge. ACS Omega, 7: 38193-38206, 2022.

Abstract - Peptides are very diverse molecules that can participate in a wide variety of biological processes. In this way, peptides are attractive for doping, since these molecules can activate or trigger biological processes that can improve the sports performance of athletes. Peptide molecules are found in the official World Anti-Doping Agency lists, mainly in sections S2, S4, and S5. In most cases, these molecules have a very short half-life in the body and/or are identical to natural molecules in the body, making it difficult to analyze them as performanceenhancing drugs. This article reviews the role of peptides in doping, with special emphasis on the peptides used as reference materials, the pretreatment of samples in biological matrices, the instrumentation, and the validation of analytical methodologies for the analysis of peptides used in doping. The growing need to characterize and quantify these molecules, especially in complex biological matrices, has generated the need to search for robust strategies that allow for obtaining sensitive and conclusive results. In this sense, strategies such as solid phase peptide synthesis (SPPS), seeking to obtain specific peptides, metabolites, or isotopically labeled analogs, is a key tool for adequate quantification of different peptide molecules in biological matrices. This, together with the use of optimal methodologies for sample pretreatment (e.g., SPE or protein precipitation), and for subsequent analysis by high-resolution techniques (mainly hyphenated LC-HRMS techniques), have become the preferred instrumentation to meet the analytical challenge involved in the analysis of peptides in complex matrices.

Sports Science Research Update

Vibration Therapy – A clinical commentary. International Journal of Sports Physical Therapy, 17(6): 984-987, 2022.

Scientific Foundation: Vibration therapy has been around for many years, and recently has regained popularity. Vibration is detected in the skin, muscles and joints by Pacinian corpuscles and travels to the Primary Somatosensory Cortex for processing via the dorsal column ascending neural pathway.1

Application Principles: It has been reported that vibration can alter stretch reflex sensitivity and in turn affect the motor unit threshold, firing rate and maximal voluntary strength on muscle action. It is believed that the effect of vibration largely depends on the duration and frequency of the stimulus provided. If applied briefly (2 – 25 seconds) vibration resulted in additional excitation of the motor neuron pool which increased initial firing rates and ultimately enhanced force production.2,3 However, if applied for prolonged periods (30+ seconds), a decrease in maximal voluntary strength occurs via presynaptic autogenic inhibition.4–6 Presynaptic autogenic inhibition is a reduction of muscle spindle activity and golgi tendon organ activation characterized by lower group Ia mean discharge rates,7 decreased reflex magnitude and increased (slow) reflex latency.8 This ultimately reduces the net excitation of the motor neuron pool and decreases maximal voluntary strength. Regarding specific application of devices, it is suggested that you work with gentle pressure over the indicated treatment area for 20 to 60 seconds. An area requiring increased treatment can be identified by experiencing a different tactile or auditory sensation. The device will either “thud” and “bounce” more aggressively over affected tissue or practitioners may even hear a different pitch in volume from the typical vibratory sound.

Indications and Contraindications: Vibration therapy may be indicated for myofascial pain, autogenic muscle inhibition and delayed onset muscle soreness. While there are no direct contraindications for its use clinicians should proceed with caution for the treatment of certain conditions. Some examples are stress fractures / reactions, neuropathy, fibromyalgia epilepsy, pregnancy, recent surgery or joint replacement, metal pins or plates, pacemakers, areas with skin rash or open wounds and in individuals with hypertension or those at risk for clotting.

Clinical Application: The ability to regulate muscle activity up or down via vibration is beneficial for clinical use and it is important that practitioners understand the scientific application to appropriately prescribe its use.

Sports Science Research Update

While there are many proposed benefits of vibration therapy the literature most strongly supports its clinical application for pain relief, improving tissue extensibility, increasing strength, and decreasing muscle soreness.

The recommendation for application is to provide gentle pressure with continuous motion into the affected tissue. You will know you found an area requiring increased attention when either the response is more pronounced ('bouncy') or louder. This does not mean that more pressure is required, simply sustained gentle pressure.